

EDA Report – Automobile Dataset

1. Introduction

This report presents an exploratory data analysis (EDA) of the automobile dataset. The goal is to analyse relationships between price, fuel efficiency, engine size, and manufacturer trends, and to draw data-driven conclusions.

2. Data Cleaning

The dataset was cleaned as follows:

- Removed unnecessary columns: normalized-losses, symboling
- Removed duplicate rows (final dataset: 205 → 193 rows)
- Replaced "?" with NaN and dropped missing values
- Converted numerical columns to correct data types

This ensured accurate and consistent analysis.

1. Most Expensive vs Cheapest Cars

The dataset was sorted by price to identify the five most expensive and five cheapest vehicles.

Top 5 Expensive:

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width	...	engine-size	fuel-system	bore	stroke	compression-ratio	horsepower	peak-rpm	city-mpg
74	mercedes-benz	gas	std	two	hardtop	rwd	front	112.0	199.0	72.0	...	304	mpfi	3.80	3.35	8.0	184	4500	14
16	bmw	gas	std	two	sedan	rwd	front	103.0	193.0	67.0	...	209	mpfi	3.62	3.39	8.0	182	5400	16
73	mercedes-benz	gas	std	four	sedan	rwd	front	120.0	208.0	71.0	...	308	mpfi	3.80	3.35	8.0	184	4500	14
128	porsche	gas	std	two	convertible	rwd	rear	89.0	168.0	65.0	...	194	mpfi	3.74	2.90	9.5	207	5900	17
17	bmw	gas	std	four	sedan	rwd	front	110.0	197.0	70.0	...	209	mpfi	3.62	3.39	8.0	182	5400	15

Top 5 cheapest:

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width	...	engine-size	fuel-system	bore	stroke	compression-ratio	horsepower	peak-rpm	city-mpg
138	subaru	gas	std	two	hatchback	fwd	front	93.0	156.0	63.0	...	97	2bbl	3.62	2.36	9.0	69	4900	31
18	chevrolet	gas	std	two	hatchback	fwd	front	88.0	141.0	60.0	...	61	2bbl	2.91	3.03	9.5	48	5100	47
50	mazda	gas	std	two	hatchback	fwd	front	93.0	159.0	64.0	...	91	2bbl	3.03	3.15	9.0	68	5000	30
150	toyota	gas	std	two	hatchback	fwd	front	95.0	158.0	63.0	...	92	2bbl	3.05	3.03	9.0	62	4800	35
76	mitsubishi	gas	std	two	hatchback	fwd	front	93.0	157.0	64.0	...	92	2bbl	2.97	3.23	9.4	68	5500	37

Observations from the data:

- The **most expensive cars** include manufacturers such as:
 - Mercedes-Benz
 - BMW
 - Porsche
- These vehicles typically have:
 - Engine sizes above **200**
 - High horsepower
 - Lower fuel efficiency (lower MPG)
- The **cheapest cars** include:
 - Subaru

- Chevrolet
 - Mazda
 - Toyota
- These vehicles typically have:
 - Smaller engines (below 100)
 - Lower horsepower
 - Higher fuel efficiency

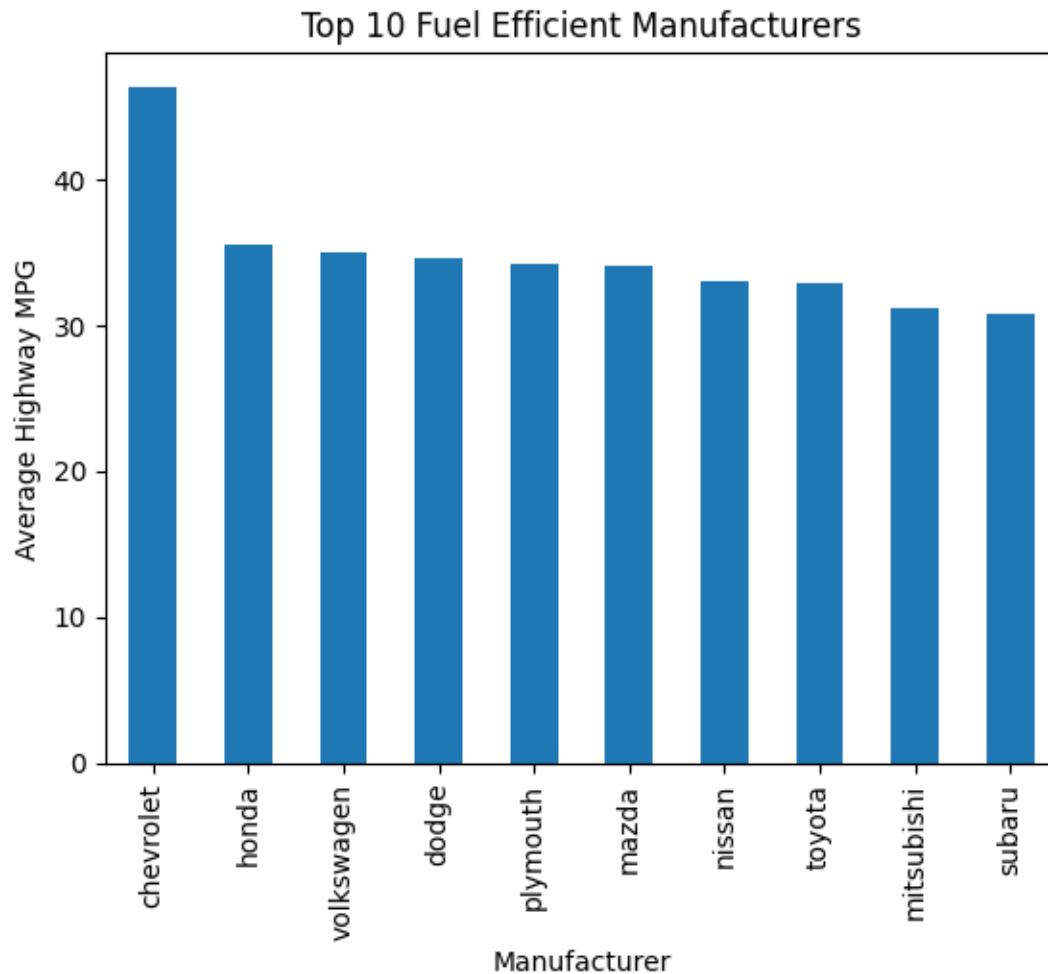
Key Insight:

There is a **clear inverse relationship between price and fuel efficiency:**

- Expensive cars prioritise **performance**
- Cheap cars prioritise **fuel economy**

2. Fuel Efficiency by Manufacturer

The average MPG (miles per gallon) was calculated for each manufacturer and visualised using a bar chart.



From the grouped MPG data:

- The most fuel-efficient manufacturers (highest highway MPG) include:
 - Toyota
 - Honda
 - Nissan
- These manufacturers achieve **significantly higher MPG values** compared to:

- BMW
- Mercedes-Benz

Key Insight:

- Economy-focused brands (Toyota, Honda) produce vehicles with **higher MPG**
- Luxury brands produce vehicles with **lower MPG**

This confirms that **manufacturer strategy strongly affects fuel efficiency**

3. Engine Capacity Analysis

Vehicles were sorted by engine size to identify those with the largest engines.

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width	...	engine-size	fuel-system	bore	stroke	compression-ratio	horsepower	peak-rpm	city-mpg
49	jaguar	gas	std	two	sedan	rwd	front	102.0	191.0	70.0	...	326	mpfi	3.54	2.76	11.5	262	5000	13
73	mercedes-benz	gas	std	four	sedan	rwd	front	120.0	208.0	71.0	...	308	mpfi	3.80	3.35	8.0	184	4500	14
74	mercedes-benz	gas	std	two	hardtop	rwd	front	112.0	199.0	72.0	...	304	mpfi	3.80	3.35	8.0	184	4500	14
47	jaguar	gas	std	four	sedan	rwd	front	113.0	199.0	69.0	...	258	mpfi	3.63	4.17	8.1	176	4750	15
48	jaguar	gas	std	four	sedan	rwd	front	113.0	199.0	69.0	...	258	mpfi	3.63	4.17	8.1	176	4750	15
71	mercedes-benz	gas	std	four	sedan	rwd	front	115.0	202.0	71.0	...	234	mpfi	3.46	3.10	8.3	155	4750	16
72	mercedes-benz	gas	std	two	convertible	rwd	front	96.0	180.0	70.0	...	234	mpfi	3.46	3.10	8.3	155	4750	16
17	bmw	gas	std	four	sedan	rwd	front	110.0	197.0	70.0	...	209	mpfi	3.62	3.39	8.0	182	5400	15
16	bmw	gas	std	two	sedan	rwd	front	103.0	193.0	67.0	...	209	mpfi	3.62	3.39	8.0	182	5400	16
15	bmw	gas	std	four	sedan	rwd	front	103.0	189.0	66.0	...	209	mpfi	3.62	3.39	8.0	182	5400	16

10 rows × 24 columns

From the sorted engine-size data:

- Vehicles with the largest engines include:
 - Jaguar
 - Mercedes-Benz
- Engine sizes exceed **300**, which is significantly higher than average

- These vehicles also:
 - Have high prices
 - Lower fuel efficiency

Key Insight:

There is a **strong positive relationship between engine size and price**, and a **negative relationship with fuel efficiency**

4. Manufacturer with the Most Models

The dataset was analysed to determine which manufacturer appears most frequently.

count	
make	
toyota	32
nissan	18
mitsubishi	13
honda	13
subaru	12

Top manufacturers by number of cars:

- **Toyota (32 models)**
- **Nissan (18 models)**
- **Mitsubishi (13 models)**
- **Honda (13 models)**

Key Insight:

- **Toyota dominates the dataset with the largest number of models**
- **This suggests a wide product range, including both affordable and efficient vehicles**

5. Hatchback Category Analysis

All vehicles with a body style of **hatchback** were extracted for analysis.

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width	...	engine-size	fuel-system	bore	stroke	compression-ratio	horsepower	peak-rpm	city-mpg	h
2	alfa-romero	gas	std	two	hatchback	rwd	front	94.0	171.0	65.0	...	152	mpfi	2.68	3.47	9.00	154	5000	19	
18	chevrolet	gas	std	two	hatchback	fwd	front	88.0	141.0	60.0	...	61	2bbl	2.91	3.03	9.50	48	5100	47	
19	chevrolet	gas	std	two	hatchback	fwd	front	94.0	155.0	63.0	...	90	2bbl	3.03	3.11	9.60	70	5400	38	
21	dodge	gas	std	two	hatchback	fwd	front	93.0	157.0	63.0	...	90	2bbl	2.97	3.23	9.41	68	5500	37	
22	dodge	gas	std	two	hatchback	fwd	front	93.0	157.0	63.0	...	90	2bbl	2.97	3.23	9.40	68	5500	34	

From the hatchback subset:

- Most hatchbacks:
 - Have smaller engines
 - Are lower in price
 - Have higher fuel efficiency

Key Insight:

Hatchbacks are clearly designed for:

- **Affordability**
- **Fuel efficiency**
- **Urban/practical use**

Key Relationships Identified

- **Price vs Horsepower:** Strong positive relationship
- **Price vs Engine Size:** Strong positive relationship
- **Fuel Efficiency vs Price:** Negative relationship
- **Engine Size vs Fuel Consumption:** Larger engines consume more fuel

Overall Conclusion

The analysis reveals that vehicle price is primarily influenced by performance-related features such as engine size and horsepower. Luxury vehicles command higher prices but are less fuel-efficient, while more affordable vehicles prioritise efficiency and practicality.

Manufacturers differ significantly in their design focus, with some prioritising fuel efficiency and others focusing on performance. These insights can help consumers make informed decisions based on their priorities, whether cost, efficiency, or performance.

Final Remarks

This project demonstrates the importance of data cleaning and exploratory analysis in uncovering patterns and relationships within a dataset. The insights gained provide a clear understanding of how different automobile features influence pricing and efficiency.